Machine Learning-Based Structural Failure Prediction and Stress Analysis Under Variable Mechanical Loading

Technical Report

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Undergraduate research project

Data provenance. All data in this report are *synthetic* and were generated from closed-form Euler–Bernoulli beam theory. No experimental measurements or finite-element simulations were used. All numerical results were produced by the accompanying code (`run_pipeline.py`, random seed 42) and are read automatically from `results/experiment_results.json` when this report is built.

Software backends used for the reported results: gradient boosting — scikit-learn `HistGradientBoostingRegressor`; neural network — scikit-learn `MLPRegressor`. XGBoost and PyTorch implementations are included in the code and are selected automatically when installed, but could not be installed in the build environment (no package-index access), so they were not executed for this report.

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Abstract

This project investigates how accurately data-driven regression and classification models can reproduce the structural response of an end-loaded rectangular cantilever when geometry, load and material properties vary simultaneously. A reproducible dataset of 30 000 physically valid beams was generated with Euler–Bernoulli theory, filtered for slenderness, small deflection and a bounded stress-to-yield ratio. Three model families—a random forest, a histogram-based gradient-boosted tree ensemble and a multi-layer perceptron (MLP)—were tuned on a validation split and evaluated once on a held-out test split of 4 500 beams. The MLP, trained on log-transformed inputs and targets, predicted maximum bending stress with $R^2 = 0.999997$ (median absolute error 0.08 %) and tip deflection with $R^2 = 0.999994$ (median error 0.09 %), clearly outperforming gradient boosting ($R^2 = 0.9964$ / 0.9906) and the random forest ($R^2 = 0.9687$ / 0.9361). For the yield criterion (factor of safety < 1) the best direct classifier reached $F_1 = 0.9877$, while thresholding the factor of safety implied by the MLP stress prediction reached $F_1 = 0.9994$, its only error occurring at a true factor of safety of 1.00024. Sensitivity sweeps show that the MLP recovers the exact power-law exponents of beam theory (e.g. $\partial \ln \sigma_{\max} / \partial \ln h = -1.999$ versus -2), whereas tree ensembles distort trends near the edges of the data and fail to extrapolate: when trained on forces up to 6574 N their median stress error on larger forces rose to 26.0 % (random forest) and 19.5 % (boosting) compared with 0.26 % for the MLP. Because the reference model is a product of powers, a linear regression in log space reproduces it exactly; this is used as a physics-informed reference that frames the limits of what the benchmark can show.

1 Introduction

Structural design repeatedly evaluates how a component responds to loads: peak stress, deformation, and the margin against yielding. For complex geometries these quantities come from numerical solvers such as the finite element method, whose cost becomes significant when thousands of evaluations are needed—in design-space exploration, optimisation, uncertainty quantification or interactive tools. *Surrogate models* replace the expensive solver with a fast approximation learned from a set of solver evaluations [3].

Before surrogates can be trusted on problems without closed-form solutions, it is instructive to study them on a problem where the exact answer is known everywhere. The cantilever beam with an end load is such a problem: its response follows from elementary beam theory [1], so every prediction can be checked exactly, both inside and outside the training domain. This project uses that setting to answer three questions:

1. How accurately do common ML regressors approximate stress and deflection for unseen beam configurations?
2. Do the learned models reproduce the *physical trends* (scaling laws) of beam theory, not just low average error?
3. How reliably can the models classify whether the yield criterion is exceeded, and where do they fail?

Scope. The project is a methodological benchmark. It does not claim novelty, and the trained models are *not* faster than the analytical formulas they imitate (Section 8.4); their value lies in what they reveal about model choice, data representation and generalisation for later work with expensive solvers.

2 Problem statement

Given the six parameters $\mathbf{x} = (L, b, h, F, E, \sigma_y)$ of a rectangular cantilever (length, width, thickness, tip force, Young's modulus, yield strength), learn mappings

$$\mathbf{x} \mapsto (\sigma_{\max},\ \delta_{\max}), \qquad \mathbf{x} \mapsto \mathbb{1}[\mathrm{FoS} < 1], \tag{1}$$

from examples, and derive the maximum strain $\varepsilon_{\max}$ and factor of safety FoS from the predicted stress. Performance is measured on beams never seen during training or model selection. All quantities are in SI units.

3 Engineering background

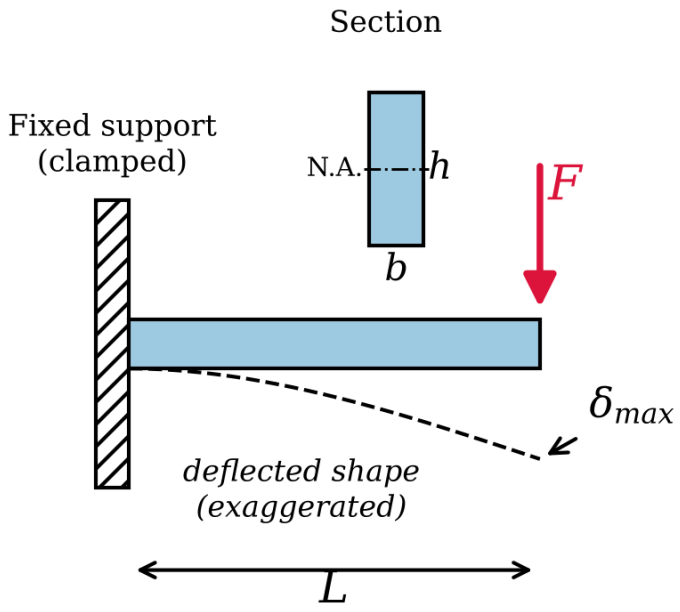


Figure 1: Problem definition: prismatic cantilever of length L and rectangular $b \times h$ section, clamped at $x = 0$, with a transverse point force F at the free end. The maximum bending moment $M = FL$ acts at the support; the maximum deflection occurs at the tip.

Euler–Bernoulli theory assumes that plane cross-sections remain plane and perpendicular to the deformed neutral axis, so the axial strain varies linearly through the depth and shear deformation is neglected [1]. The theory is accurate for slender beams (span-to-depth ratio of roughly ten or more) under small deflections. The assumptions adopted throughout are:

- linear-elastic, homogeneous, isotropic material (Hooke's law, $\sigma = E\varepsilon$);
- small deflections and rotations (geometrically linear);
- static load applied at the tip, no self-weight;
- prismatic rectangular cross-section bending about its strong or weak axis as defined by h;
- no stress concentrations, residual stresses, or support compliance;
- failure is represented only by the onset of yield at the most stressed fibre.

The yield check compares the nominal elastic stress with the yield strength, the usual first design check for ductile members under static load [2]. It does *not* address fatigue, buckling (including lateral–torsional buckling of tall narrow sections), fracture, plastic collapse or dynamic effects.

4 Mathematical formulation

With x measured from the fixed end, the bending moment and second moment of area are

$$M(x) = F\,(L - x), \qquad I = \frac{bh^3}{12}. \tag{2}$$

The flexure formula gives the axial stress at height y above the neutral axis,

$$\sigma_{xx}(x, y) = \frac{M(x)\,y}{I}, \tag{3}$$

which is largest at the support ($x = 0$) on the outer fibres ($|y| = h/2$):

$$\boxed{\sigma_{\max} = \frac{FL\,(h/2)}{bh^3/12} = \frac{6FL}{bh^2}} \tag{4}$$

Integrating $EI\,w'' = M$ with $w(0) = w'(0) = 0$ gives the elastic curve and tip deflection

$$w(x) = \frac{Fx^2(3L - x)}{6EI}, \qquad \boxed{\delta_{\max} = w(L) = \frac{FL^3}{3EI} = \frac{4FL^3}{Ebh^3}} \tag{5}$$

The maximum strain, factor of safety and yield label follow as

$$\varepsilon_{\max} = \frac{\sigma_{\max}}{E}, \qquad \mathrm{FoS} = \frac{\sigma_y}{\sigma_{\max}}, \qquad y_{\mathrm{cls}} = \begin{cases} 0 & \mathrm{FoS} \geq 1 \text{ (safe under the adopted criterion)} \\ 1 & \mathrm{FoS} < 1 \text{ (yield criterion exceeded)} \end{cases} \tag{6}$$

Power-law structure. Equations (4)–(5) are products of powers of the inputs. Taking logarithms makes them exactly linear:

$$\log \sigma_{\max} = \log 6 + \log F + \log L - \log b - 2 \log h, \tag{7}$$

$$\log \delta_{\max} = \log 4 + \log F + 3 \log L - \log E - \log b - 3 \log h. \tag{8}$$

This fact drives two design decisions: models are trained on $\log_{10}$ targets, and a linear least-squares fit in log space is included as a physics-informed reference model. The exponents also give exact *elasticities* $\partial \ln y / \partial \ln x_i$ against which learned sensitivities are tested (Section 10).

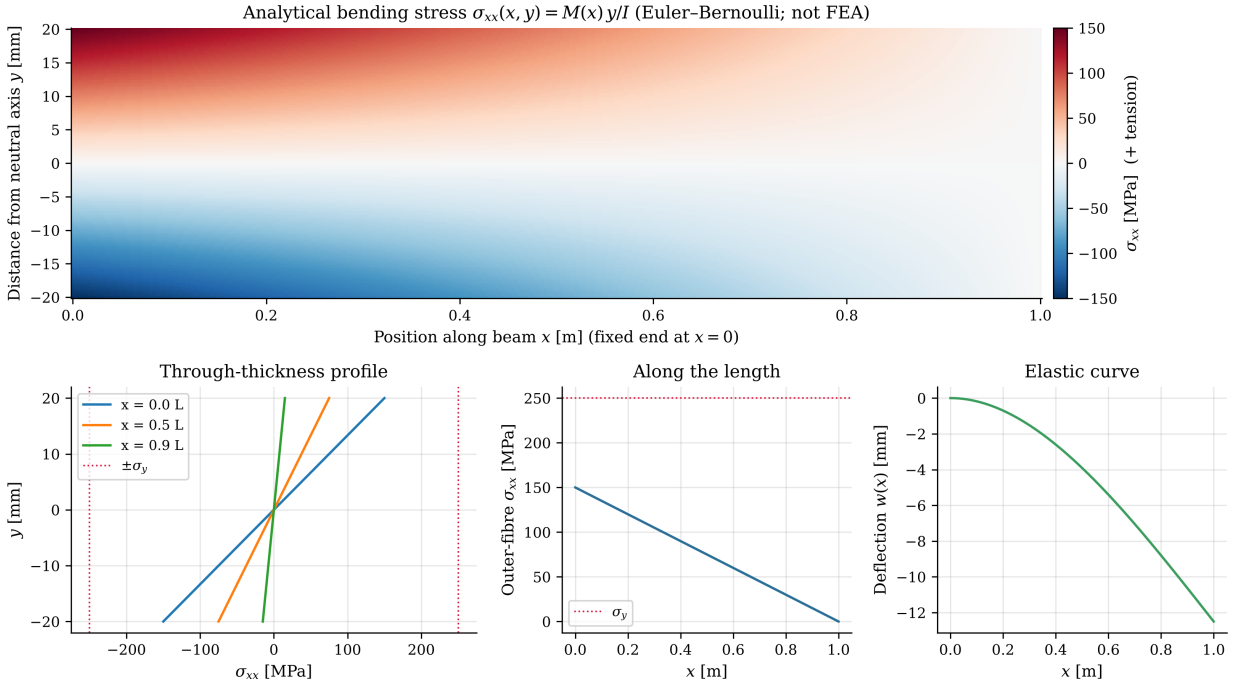


Figure 2: Analytical fields for the baseline beam ($L = 1\,\text{m}$, $b = 50\,\text{mm}$, $h = 40\,\text{mm}$, $F = 2\,\text{kN}$, $E = 200\,\text{GPa}$, $\sigma_y = 250\,\text{MPa}$): bending stress over the beam (top), through-thickness profiles, outer-fibre stress along the span, and the elastic curve. Peak stress is 150 MPa (FoS = 1.67) and tip deflection 12.5 mm. These are closed-form beam-theory fields, **not** finite-element results; local effects near the clamp are not represented.

5 Dataset generation

5.1 Sampling and physical-validity constraints

The six inputs were sampled independently from uniform distributions (Table 1) using NumPy's `default_rng` with seed 42 [15]. Each candidate was evaluated with the analytical solver and kept only if it satisfied three validity constraints:

1. $L/h \geq 10$: slender beam, consistent with neglecting shear deformation;
2. $\delta_{\max}/L \leq 0.05$: small-deflection assumption;
3. $\sigma_{\max}/\sigma_y \leq 3$: the nominal elastic stress is retained only as a screening quantity near the yield boundary—beams loaded far into the plastic range, where an elastic stress has no physical meaning, are excluded.

Of 120 000 candidates, 34 085 (28.4 %) passed; the first 30 000 form the dataset. Automatic checks confirm that there are no missing or non-finite values, all rows satisfy the constraints, all targets reproduce the solver to a relative tolerance of 10^{-12}, and there are no duplicate inputs.

Table 1: Input parameters, sampling ranges and resulting dataset statistics.

Parameter	Symbol	Unit	Sampling range	Mean	Median
Length	L	m	0.5–2	1.01	0.919
Width	b	mm	20–100	66.5	68.9
Thickness	h	mm	10–50	38.8	40.4
Force	F	N	100–10000	3.31e+03	2.61e+03
Young's modulus	E	GPa	60–210	145	149
Yield strength	σ_y	MPa	150–500	325	326

5.2 Exploratory data analysis

The rejection step makes the input distributions non-uniform (Figure 3): long beams, thin beams, large forces and low moduli are under-represented, because these combinations most often violate the deflection or stress limits. It also introduces weak correlations between nominally independent inputs (e.g. Pearson $r(h, F) = 0.31$, $r(L, F) = -0.34$; Figure 5). 19.0 % of beams exceed the yield criterion, giving a moderately imbalanced classification problem. Responses span two to three orders of magnitude (Figure 4). IQR-rule "outliers" in the log-transformed targets (763 for stress, 685 for deflection) are genuine extreme configurations rather than errors and were retained.

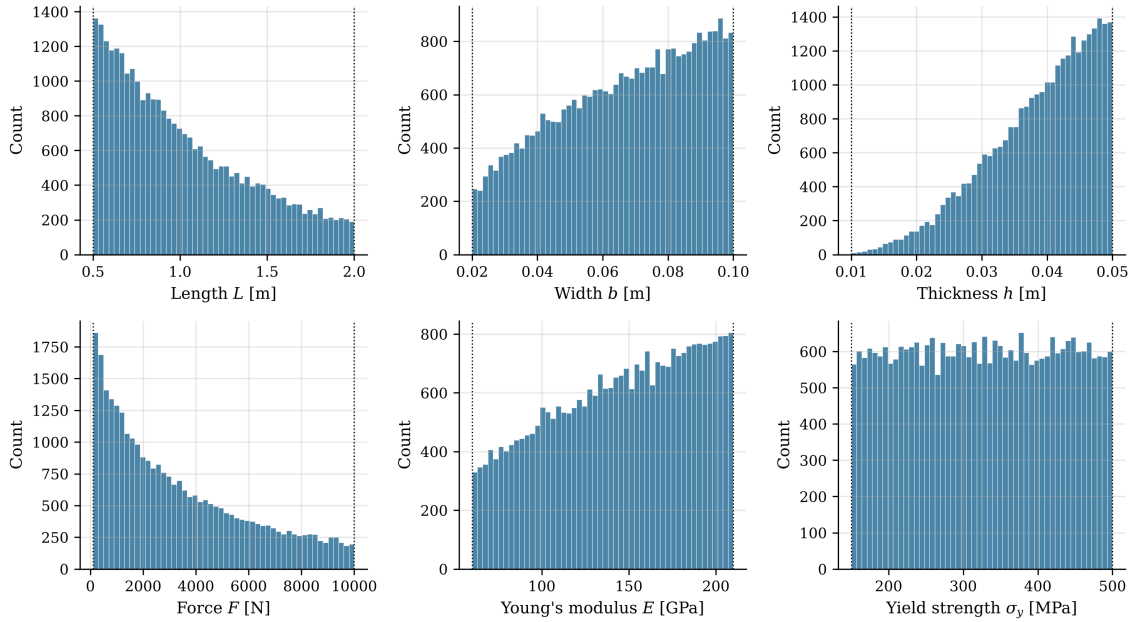


Figure 3: Input distributions after rejection sampling; dotted lines are the sampling bounds.

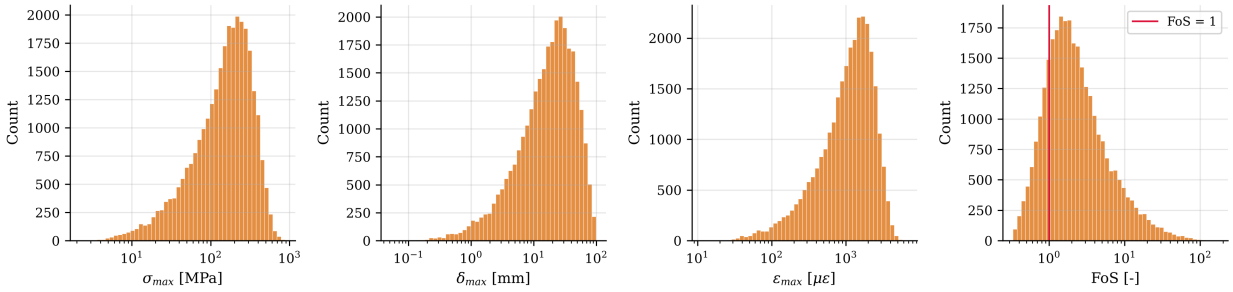


Figure 4: Distributions of the analytical responses (log axes). The red line marks FoS = 1.

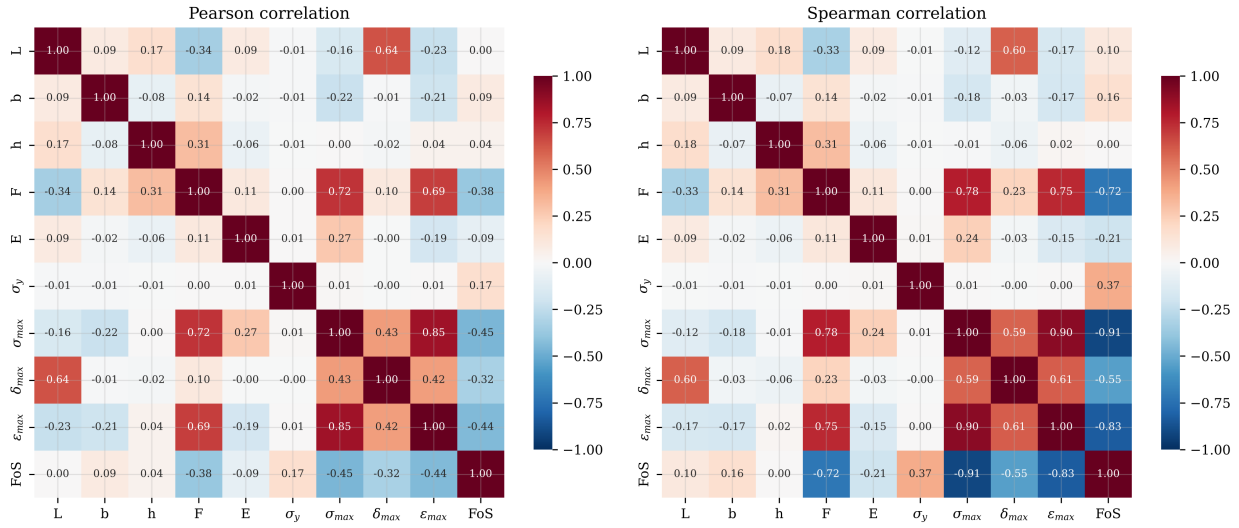


Figure 5: Pearson and Spearman correlation matrices of inputs and responses. Force is the input most strongly correlated with stress; length with deflection.

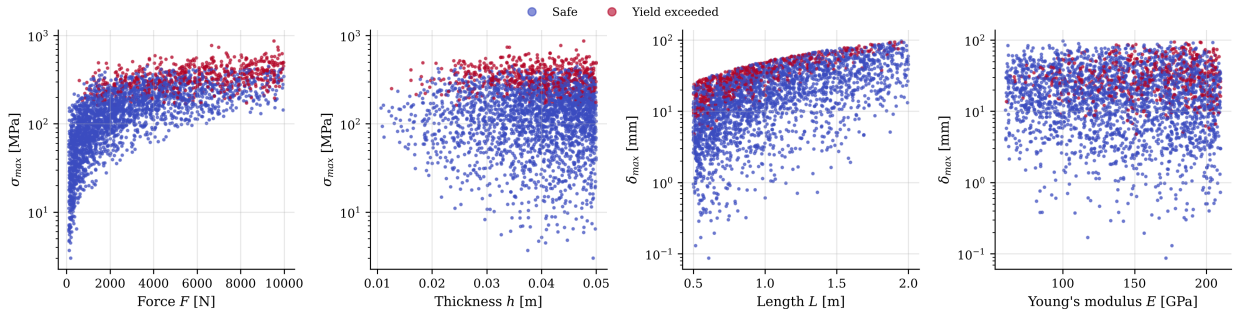


Figure 6: Marginal input–response relationships for 3 000 random samples (colour: yield class). The wide scatter at fixed input shows that each response depends jointly on several parameters.

6 Machine learning methodology

6.1 Data representation

Stress and deflection are strictly positive and span several decades, so all regressors learn $\log_{10}\sigma_{\max}$ and $\log_{10}\delta_{\max}$; predictions are transformed back to SI units before any metric is computed. Strain and FoS are *derived* from the predicted stress and the known inputs ($\varepsilon = \hat{\sigma}/E$, $\widehat{\mathrm{FoS}} = \sigma_y/\hat{\sigma}$) rather than learned separately, which keeps the four outputs mutually consistent.

6.2 Models

Random forest [4]. Bagged ensemble of deep regression trees with a two-output target. Tree ensembles are invariant to monotone transformations of inputs, so raw SI features are used.

Gradient-boosted trees [5]. scikit-learn `HistGradientBoostingRegressor`: an additive ensemble of shallow trees fitted sequentially to the residuals, with histogram-binned features (the approach popularised by LightGBM [7]) and early stopping on held-out data. One model per target.

Multi-layer perceptron [8]. scikit-learn `MLPRegressor` [11]: fully connected network with hidden layers (256, 256, 128), ReLU activations, Adam optimiser [9] (learning rate 0.001), mini-batches of 256, L_2 penalty 10^{-5} and early stopping on a validation fraction. Inputs are

$\log_{10}$-transformed and standardised with statistics from the training split only; targets are standardised log values.

Power-law reference. Ordinary least squares on $\log_{10}$ inputs. Because the true model is exactly linear in this space, this reference is expected to be exact; it is reported to make that explicit, not as a competing "ML" method.

MLP ablation (raw inputs). Identical MLP trained on linearly standardised (not log-transformed) inputs, to measure the value of the log representation.

For classification of the yield criterion, a random-forest classifier, a gradient-boosted classifier and an MLP classifier were trained directly on the six inputs. In addition, a *derived* classifier thresholds the FoS implied by each stress regressor.

6.3 Hyper-parameter selection

A small grid was searched for each model family, selecting on the validation-set RMSE of the $\log_{10}$ targets (Table 2). The test split was not used for any choice.

Table 2: Hyper-parameter grids and selected values (validation RMSE in $\log_{10}$ units, averaged over both targets).

Model	Grid searched	Selected (validation RMSE)
Random forest	max_features $\in \{1.0, 0.6\}$, min_samples_leaf $\in \{2, 4\}$; 100 trees	max_features=1.0, min_samples_leaf=2 (0.0617)
HistGradBoost	learning_rate $\in \{0.05, 0.1\}$, max_leaf_nodes $\in \{31, 63\}$; up to 2000 iterations, early stopping (50)	learning_rate=0.05, max_leaf_nodes=31 (0.0253)
MLP (sklearn)	hidden layers $\in \{(128, 128, 64), (256, 256, 128)\}$; Adam, lr 10^{-3}	hidden=(256, 256, 128), lr=0.001 (0.0006)

7 Experimental setup

- **Split:** 70/15/15 % train/validation/test (21 000/4 500/4 500 beams), stratified on the yield label, seed 42. Feature scaling and target scaling are fitted on the training split only.
- **Regression metrics:** MAE, RMSE and R^2 in SI units, plus absolute percentage error (APE) statistics (mean, median, 95th percentile, maximum), which are more informative for targets spanning several decades.
- **Classification metrics:** accuracy, precision, recall, F_1 (positive class = yield exceeded), ROC-AUC and confusion matrices.
- **Extrapolation test:** models retrained on beams with $F \leq 6574\,\mathrm{N}$ (85th percentile of F; 25 500 beams) and evaluated on the 4 500 beams with larger force.
- **Sensitivity analysis:** one-at-a-time sweeps around the baseline beam and global variance shares (Section 10).
- **Reproducibility:** every stage is scripted (scikit-learn [11], Matplotlib [16]); `python run_pipeline.py` regenerates the dataset, models, metrics and figures, then runs 11 automated tests including hand-calculated solver checks and a test that re-scores reloaded models and compares the result with the saved metrics. Two full runs from an empty directory produced the same dataset and models; with single-threaded forest inference, repeated evaluations give bit-identical metrics. Compute: 2 CPU cores, no GPU; the timed regression-training stages took 245 s in total.

8 Model evaluation

8.1 Regression

Table 3: Test-set regression metrics ($N = 4\,500$). MAE and RMSE are in MPa for stress and mm for deflection. APE: absolute percentage error. Best ML model per column in bold.

Target	Model	MAE	RMSE	R^2	Median APE %	P95 APE %	Max APE %
Stress	Random Forest	13.7	22.68	0.9687	6.081	22.04	135.2
	HistGradBoost	4.789	7.659	0.9964	2.184	7.709	38.56
	MLP (sklearn)	**0.1582**	**0.2352**	**0.999997**	**0.08017**	**0.2659**	**0.6217**
	MLP (raw inputs)	1.943	2.992	0.9995	0.8817	4.903	32.19
	Power-law (log-OLS)	7.13e-13	9.98e-13	1.000000	4.17e-13	9.03e-13	1.50e-12
Deflection	Random Forest	2.996	4.969	0.9361	10.61	35.83	146.4
	HistGradBoost	1.138	1.909	0.9906	4.001	14.31	96.73
	MLP (sklearn)	**0.02872**	**0.04867**	**0.999994**	**0.09494**	**0.294**	**0.6887**
	MLP (raw inputs)	0.3546	0.7046	0.9987	1.196	5.155	32.33
	Power-law (log-OLS)	2.00e-13	3.39e-13	1.000000	6.78e-13	1.99e-12	2.32e-11

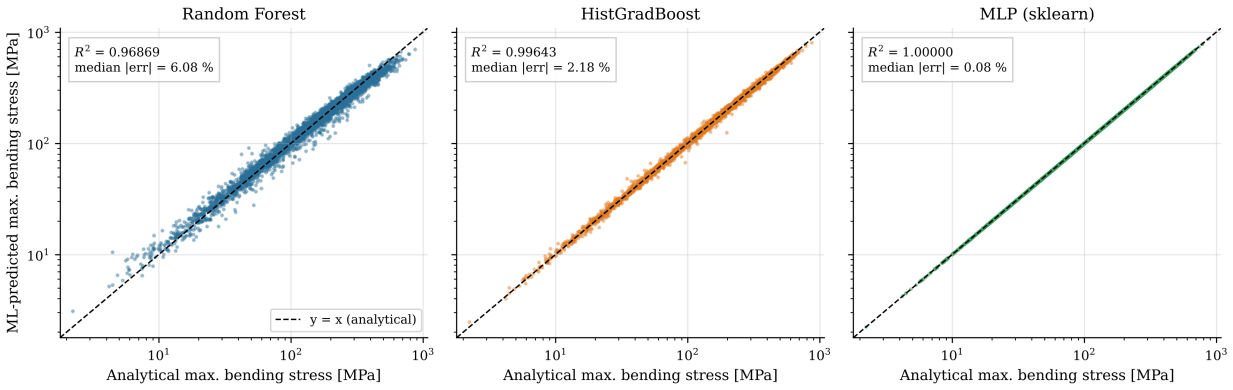


Figure 7: Predicted versus analytical maximum bending stress on the test set (log–log axes).

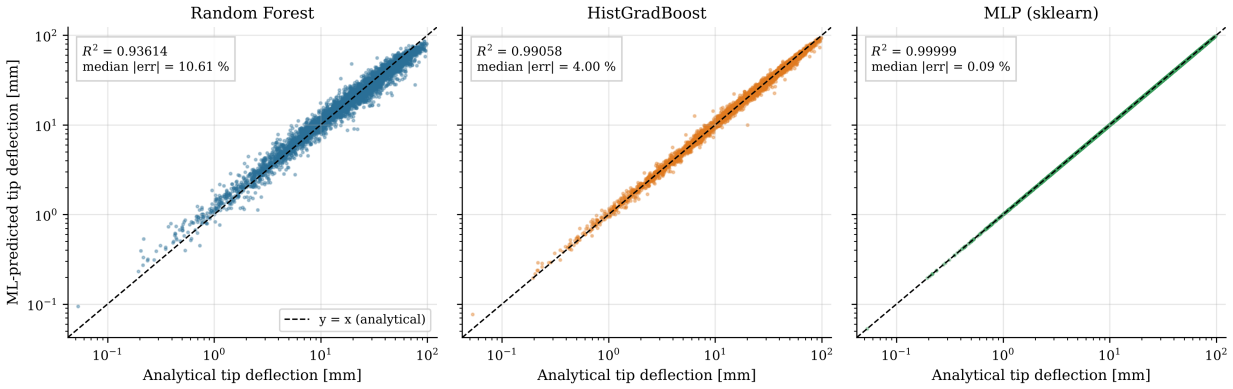


Figure 8: Predicted versus analytical tip deflection on the test set (log–log axes).

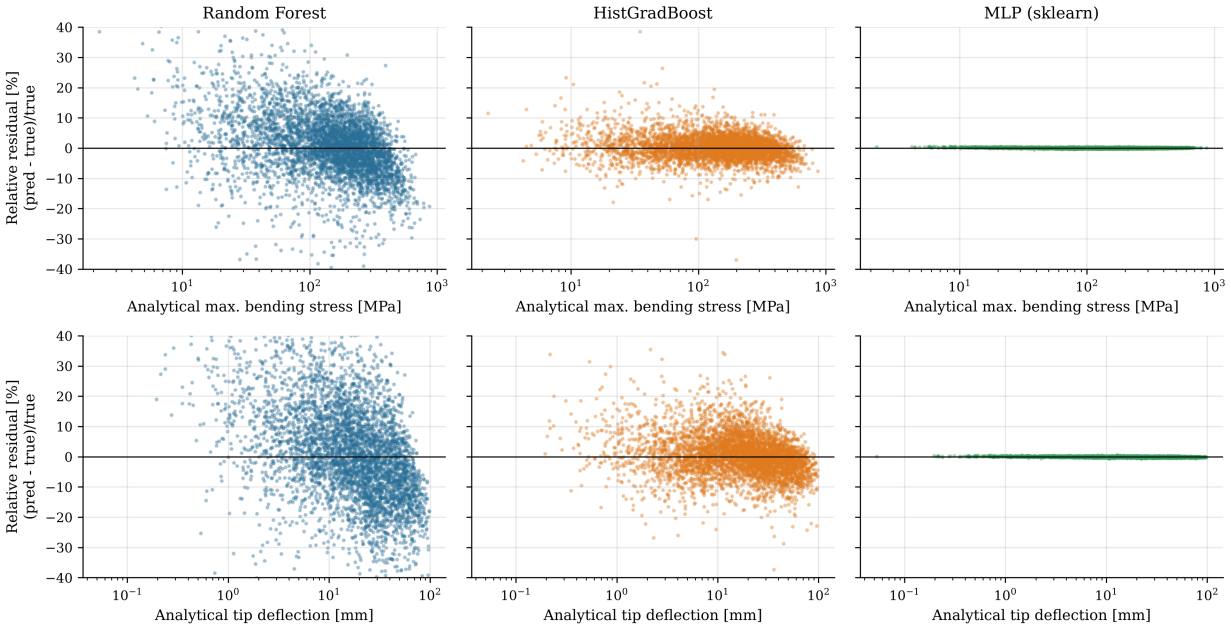


Figure 9: Relative residuals versus analytical value (axes clipped at $\pm 40\,\%$). Tree-ensemble errors are largest at the extremes of the target range, where training data are sparse.

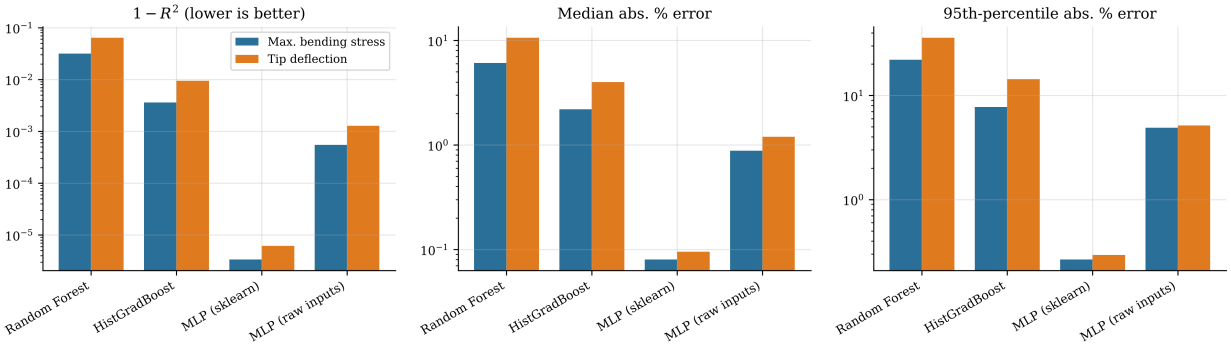


Figure 10: Test-set comparison of regressors (log scale). The power-law reference is omitted because its errors are at machine precision (median APE $\sim 10^{-12}\,\%$).

8.2 Yield classification

Table 4: Test-set yield-classification metrics (positive class: FoS < 1; 854 positives of 4 500).

Classifier	Accuracy	Precision	Recall	F_1	ROC-AUC
Random Forest (direct)	0.9522	0.9323	0.8068	0.8650	0.9873
HistGradBoost (direct)	0.9722	0.9451	0.9063	0.9253	0.9964
MLP (sklearn) (direct)	0.9953	0.9906	0.9848	0.9877	0.9999
FoS from Random Forest stress	0.9684	0.9484	0.8817	0.9138	0.9949
FoS from HistGradBoost stress	0.9904	0.9810	0.9684	0.9747	0.9995
FoS from MLP (sklearn) stress	0.9998	0.9988	1.0000	0.9994	1.0000
FoS from MLP (raw inputs) stress	0.9949	0.9906	0.9824	0.9865	0.9998
FoS from power-law reference	1.0000	1.0000	1.0000	1.0000	1.0000

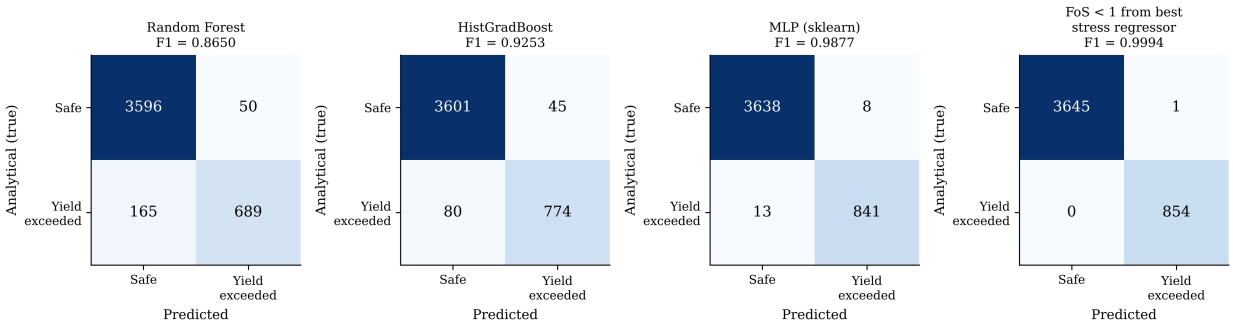


Figure 11: Confusion matrices: three direct classifiers and the classifier derived from the best stress regressor.

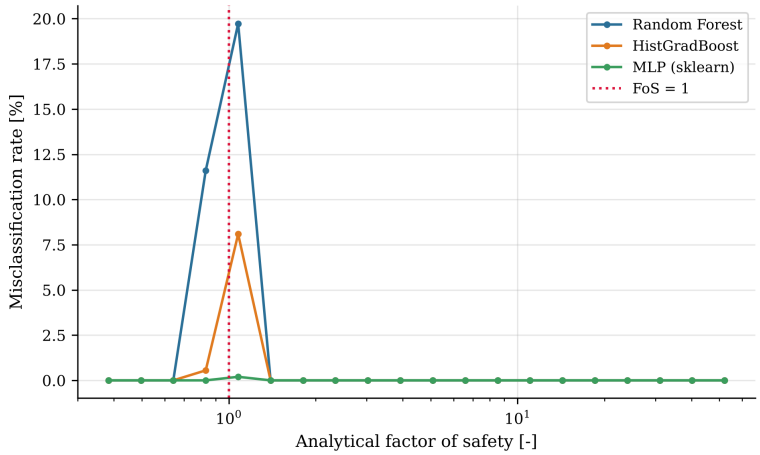


Figure 12: Misclassification rate of the regression-derived yield label, binned by true FoS.

8.3 Extrapolation beyond the training range

Table 5: Extrapolation test: trained on $F \leq 6574\,\mathrm{N}$, tested on larger forces (4 500 beams).

Model	Stress R^2	Stress Median APE %	Deflection R^2	Deflection Median APE %
Random Forest	0.2877	26.04	0.7544	24.22
HistGradBoost	0.5843	19.54	0.8402	19.23
MLP (sklearn)	0.9999	0.26	1.0000	0.20
Power-law (log-OLS)	1.0000	0.00	1.0000	0.00

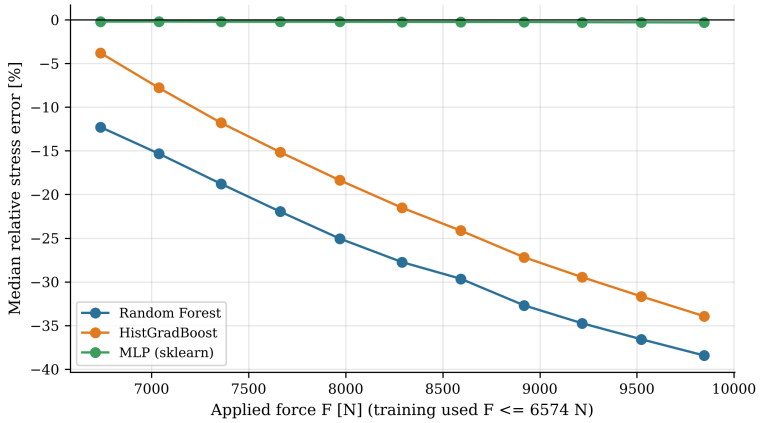


Figure 13: Median relative stress error versus applied force in the extrapolation test. Tree ensembles underpredict increasingly as F moves beyond the training range.

8.4 Computational cost

Predicting the 4 500 test beams took 0.2 ms with the vectorised analytical solver, versus 23.4 ms (MLP), 103.0 ms (random forest) and 401.6 ms (gradient boosting) on the same machine. As expected, a surrogate is *slower* than the closed-form expression it imitates; the timing is reported to avoid any implied speed-up claim.

9 Results and discussion

Ranking. For every regression target the MLP is the best learned model by a wide margin: its median error is roughly 27 times smaller than gradient boosting and 76 times smaller than the random forest for stress (Table 3). Gradient boosting is consistently better than the random forest, which is the weakest model on every metric.

Why the MLP wins. In log space the target is a linear function of the log inputs. An MLP fed log-standardised inputs only has to learn a nearly linear map, which it does to within ∼0.1 %. The ablation confirms that the representation, not only the architecture, matters: the same MLP on raw standardised inputs has a median stress error of 0.88 %, about 11 times larger, because it must synthesise products and inverse powers such as $1/h^2$ from additive units.

Why tree ensembles struggle. Trees [12] approximate a function by axis-aligned, piecewise-constant pieces. A smooth multiplicative function of six variables needs very many such pieces, and each leaf averages training targets, which biases predictions towards the mean at the edges of the data. This is visible as heteroscedastic residuals that grow at the extremes of the target range (Figure 9) and as a maximum stress error of 135 % for the random forest. Boosting reduces the bias by fitting residuals sequentially [5], explaining its better scores. The random forest is also hampered by averaging fully grown trees on only 21 000 samples in a six-dimensional space.

Classification. Direct classifiers must learn the curved yield boundary $\sigma_y b h^2 = 6FL$ from labels alone, and their errors are concentrated close to it. Deriving the label from an accurate stress regressor is more effective: the MLP-derived classifier made a single error out of 4 500 (at true FoS = 1.00024), while the best direct classifier (MLP) made 21. Figure 12 shows that misclassification happens only in a narrow band around FoS = 1, where a small relative error in stress flips the label. For safety-related use, a conservative margin (e.g. flagging $\widehat{\mathrm{FoS}} < 1 + k\cdot$ expected error) would be required; recall (missed yield cases) is the more critical error.

Extrapolation. When the test forces exceed the training range, tree ensembles cannot predict stresses above the largest training targets in the corresponding leaves, so the random forest's stress R^2 drops to 0.288 (Table 5). The log-input MLP stays accurate because the learned near-linear log map continues smoothly. This is the most practically relevant difference between the families: test-set accuracy inside the training distribution does not guarantee sensible behaviour outside it.

10 Sensitivity analysis

10.1 One-at-a-time sweeps

Following standard practice [13], each input was swept across its full range with the others fixed at the baseline (Figure 14). Parts of a sweep that violate the validity constraints (grey) lie outside the training data. The local log–log slope of each model's prediction over the valid part of each sweep was compared with the exact exponent (Table 6, Figure 15).

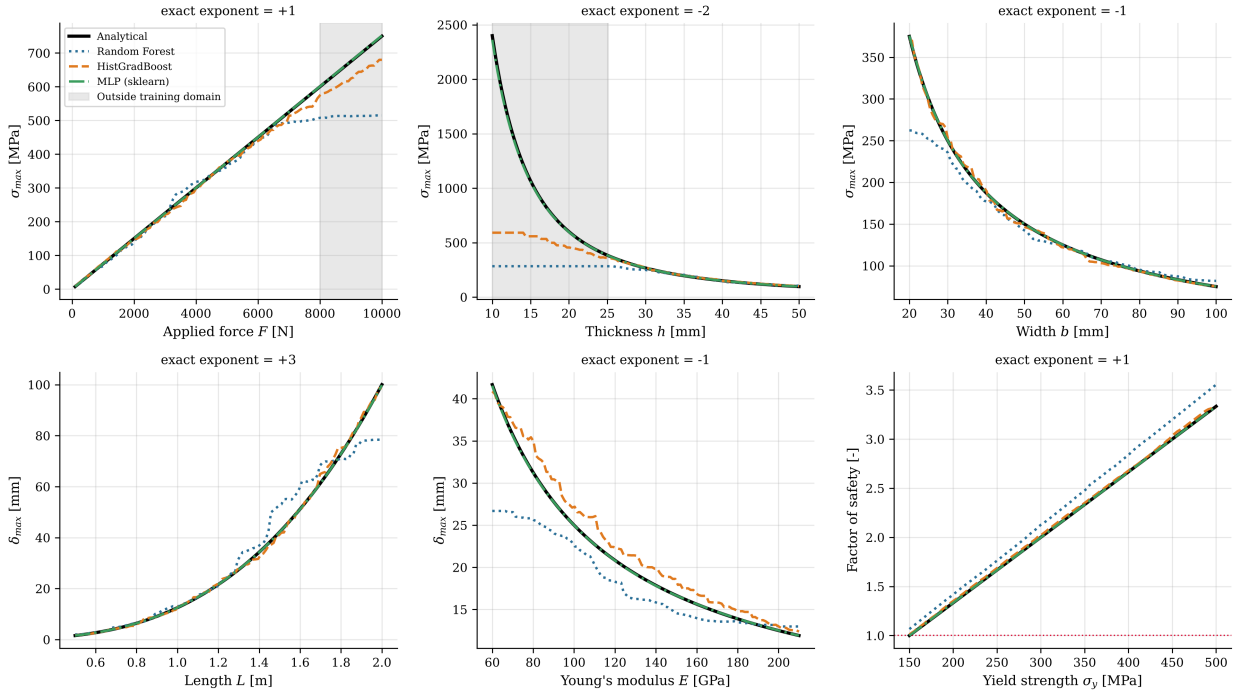


Figure 14: One-at-a-time sweeps around the baseline beam. Black: analytical; coloured: ML models. Grey: outside the training domain.

Table 6: Elasticities $\partial \ln y / \partial \ln x$ recovered from the sweeps (valid domain only) and fraction of sweep steps with the physically correct sign of change.

Effect	Exact	Random forest	Boosting	MLP	Correct-sign steps (RF / GB / MLP)
$F \to \sigma_{\max}$	+1	+0.976	+0.970	+1.000	98% / 94% / 100%
$h \to \sigma_{\max}$	-2	-1.663	-1.910	-1.999	93% / 92% / 100%
$b \to \sigma_{\max}$	-1	-0.833	-1.011	-1.001	91% / 87% / 100%
$L \to \delta_{\max}$	+3	+2.946	+3.017	+2.998	89% / 92% / 100%
$E \to \delta_{\max}$	-1	-0.736	-0.997	-0.997	70% / 77% / 100%
$\sigma_y \to$ FoS	+1	+1.003	+0.995	+1.000	100% / 100% / 100%

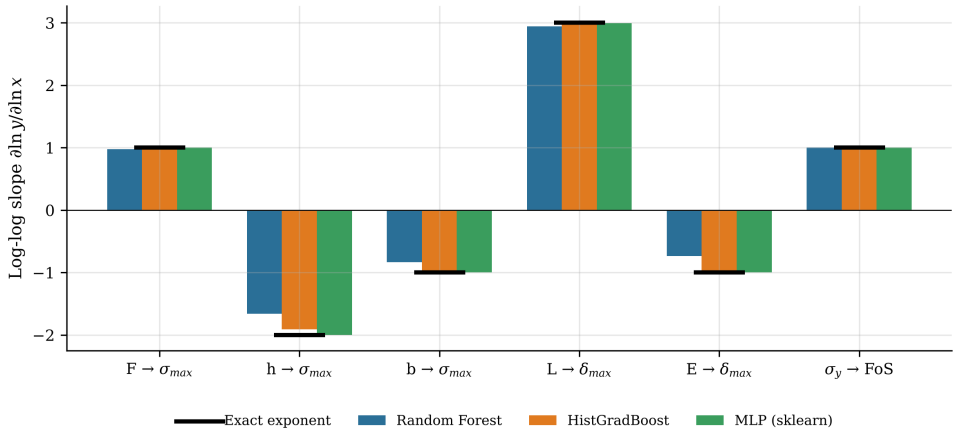


Figure 15: Elasticities recovered by each model compared with the exact exponents.

10.2 Physical interpretation

- **Force → stress (exponent +1).** Stress is proportional to load; doubling F doubles $\sigma_{\max}$. All models capture this inside the domain; beyond the training range the tree models flatten (Figure 14, top left).
- **Thickness → stress (−2).** The section modulus $bh^2/6$ grows with h^2, so thickness is the most efficient way to reduce stress: a 25 % thicker beam lowers stress by 36 %. Trees recover a slope of only -1.66 (random forest) because thin beams are rare in the filtered data.
- **Width → stress (−1).** Width enters linearly in I and in the section modulus, hence the weaker effect.
- **Length → deflection (+3).** The cubic dependence makes span the dominant driver of deflection; doubling L increases $\delta_{\max}$ eightfold.
- **Young's modulus → deflection (−1).** Stiffer material reduces deflection proportionally but, for this statically determinate beam, leaves the stress unchanged. The random forest's slope of -0.74 shows it partly confuses the role of E, which carries little variance in the data.
- **Yield strength → FoS (+1).** Since σ_y does not affect stress, FoS scales linearly with it; the yield boundary in Figure 17 follows $h_{\text{crit}} = \sqrt{6FL/(b\sigma_y)}$.

10.3 Global sensitivity over the dataset

Because $\log y = \sum_i a_i \log x_i + c$ exactly, the variance of $\log y$ can be split into contributions $S_i = a_i \operatorname{Cov}(\log x_i, \log y)/\operatorname{Var}(\log y)$, which sum to one even for correlated inputs [14]. Over the dataset, force accounts for 97 % of the log-variance of stress, while thickness contributes only 0.2 % despite its exponent of -2 (Figure 16). This is a property of the *filtered dataset*, not of beam theory: thickness varies over a narrow log-range after filtering and is positively correlated with force, so thick beams tend to carry large loads and the two effects partly cancel. For deflection, length (59 %) and force (32 %) dominate.

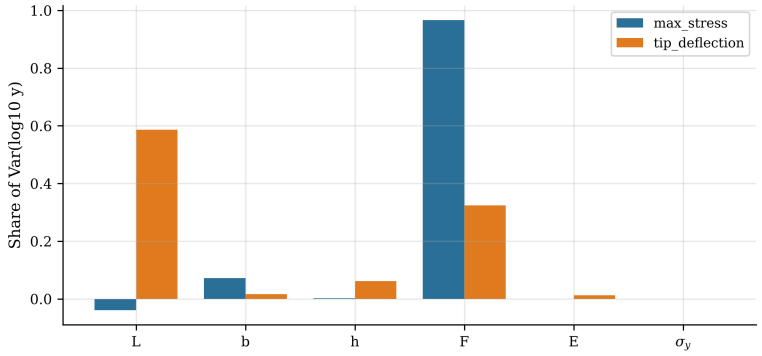


Figure 16: Covariance-based shares of $\mathrm{Var}(\log_{10} y)$ by input over the dataset.

11 Engineering interpretation

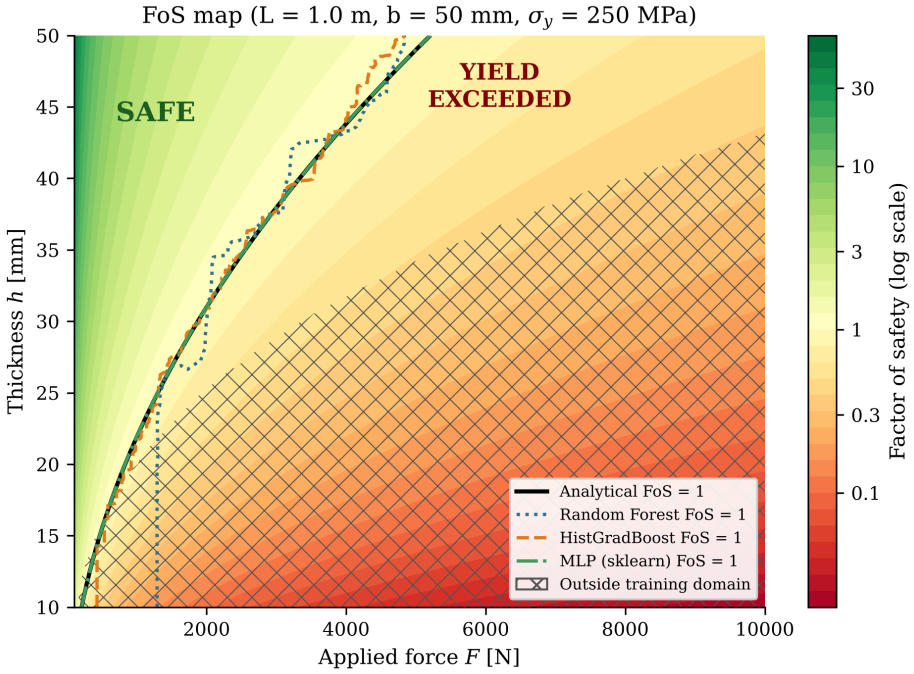


Figure 17: Factor-of-safety map over force and thickness (other parameters at baseline). Black: analytical yield boundary; coloured: boundaries implied by each ML stress model. Hatched: outside the training domain.

Figure 17 translates the accuracy figures into design terms. The MLP boundary is indistinguishable from the analytical curve, while the random-forest boundary is a staircase and, where it enters the hatched region, extends vertically instead of following the curve—a direct visual consequence of piecewise-constant extrapolation. A designer using the random forest near $h \approx 25$–35 mm could misjudge the required thickness by a few millimetres. The practical lessons are:

1. encode known physics in the representation (log transforms, dimensionless groups) where possible;
2. test learned models for physically correct trends and behaviour at and beyond the data boundary, not only for average test error;
3. treat classifications near FoS = 1 with an explicit safety margin, and prefer conservative errors.

12 Limitations

- **Synthetic, noise-free data.** Targets come from a closed-form solution with no measurement noise, modelling error or material scatter. Real experimental or FEA data would be noisier and much scarcer.
- **Simplified mechanics.** Linear-elastic Euler–Bernoulli theory with a nominal yield check; no shear deformation, stress concentration at the clamp, plasticity, buckling, fatigue or fracture. "Failure" means only that the elastic stress reaches yield.
- **Trivially identifiable target.** A log-linear regression is exact, so the benchmark measures how well generic learners recover a known power law, not their ability to discover unknown physics.
- **Sampling choices.** Uniform sampling with rejection produces non-uniform, correlated inputs; independent (E, σ_y) sampling does not reflect real alloy pairings. Results depend on these choices.
- **Software backends.** The project specification called for XGBoost and PyTorch. These packages could not be installed in the build environment, so the reported results use scikit-learn's histogram gradient boosting (same algorithm family as XGBoost) and `MLPRegressor`. The XGBoost/PyTorch code paths exist but are untested; their numbers would differ slightly from those reported here.
- **Limited tuning.** Small hyper-parameter grids were used; tree models might improve with larger ensembles, but cannot overcome their extrapolation behaviour.

13 Future scope

- Replace the analytical solver with FEA (e.g. 2-D plane-stress or 3-D solid models) to include stress concentrations at the support, and study accuracy versus number of expensive simulations (learning curves, active learning).
- Extend to load cases and geometries without closed-form solutions: distributed loads, I-sections, plates with holes.
- Predict full stress or displacement fields with convolutional or graph neural networks instead of scalar maxima.
- Physics-informed training (dimensionless inputs, monotonicity constraints, PINN-style residual losses).
- Add material nonlinearity (elasto-plastic response) and noisy labels; quantify predictive uncertainty (ensembles, quantile regression) to set rational classification margins.

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A Reproducing the results

```
pip install -r requirements.txt
python run_pipeline.py           # full pipeline + tests
python run_pipeline.py --report  # also rebuild this PDF
streamlit run app.py             # dashboard
```

All numbers in this report are inserted automatically from `results/experiment_results.json`, `results/sensitivity_results.json`, `models/*/config.json` and `data/generation_stats.json` by `report/build_report.py`.